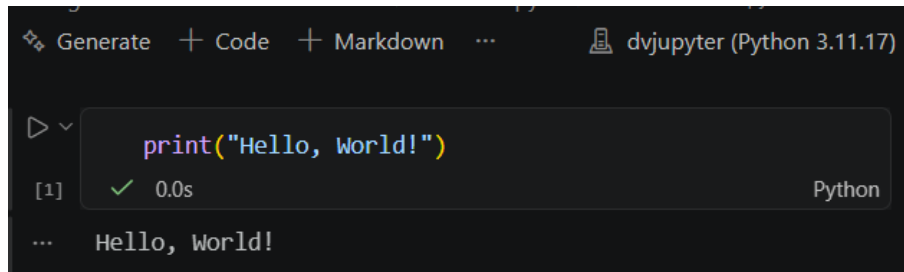


<https://github.com/DanielStein15/assignment-1-dms22e-DanielStein>

Setup and data collection:

I completed the assignment using Python 3.11 in the dvjupyter Conda environment with Jupyter in VSCode. I used OpenAlex API to collect FSU-affiliated works from 2021 - 2026, using cursor-based pagination, downloading 27733 works. The JSONL file was left out. I didnt have any issues with the API.

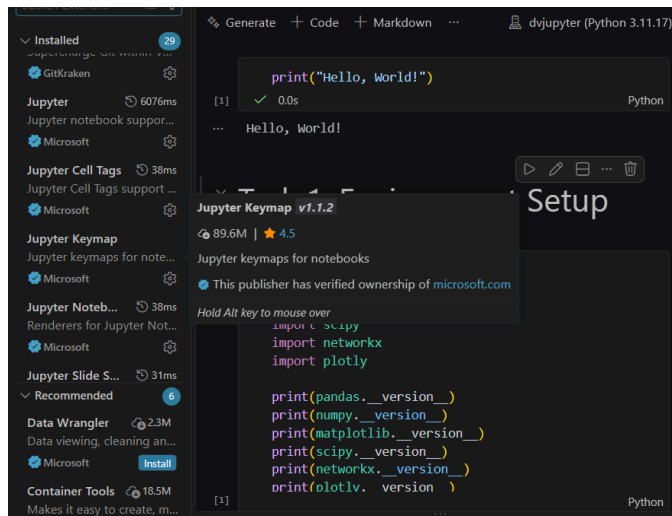


```
Generate + Code + Markdown ... dvjupyter (Python 3.11.17)
```

```
[1] ✓ 0.0s Python
```

```
print("Hello, World!")
```

```
... Hello, World!
```



```
Generate + Code + Markdown ... dvjupyter (Python 3.11.17)
```

```
[1] ✓ 0.0s Python
```

```
print("Hello, World!")
```

```
... Hello, World!
```

```
Jupyter Keymap v7.7.2
```

```
89.6M | 4.5
```

```
Jupyter keymaps for notebooks
```

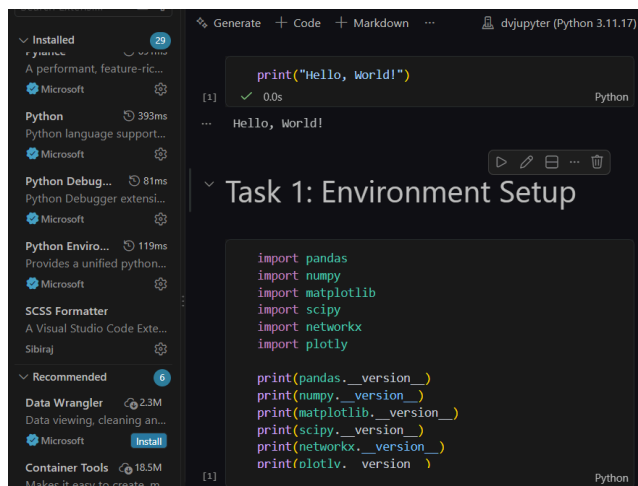
```
This publisher has verified ownership of microsoft.com
```

```
Hold Alt key to mouse over
```

```
import pandas
import numpy
import matplotlib
import scipy
import networkx
import plotly
```

```
print(pandas.__version__)
print(numpy.__version__)
print(matplotlib.__version__)
print(scipy.__version__)
print(networkx.__version__)
print(plotly.__version__)
```

```
[1] Python
```



```
Generate + Code + Markdown ... dvjupyter (Python 3.11.17)
```

```
[1] ✓ 0.0s Python
```

```
print("Hello, World!")
```

```
... Hello, World!
```

```
Task 1: Environment Setup
```

```
import pandas
import numpy
import matplotlib
import scipy
import networkx
import plotly
```

```
print(pandas.__version__)
print(numpy.__version__)
print(matplotlib.__version__)
print(scipy.__version__)
print(networkx.__version__)
print(plotly.__version__)
```

```
[1] Python
```

Data cleaning:

I flattened each OpenAlex work into one row with these attributes: openalex_id, doi, title, publication_year, publication_date, type, cited_by_count, venue, venue_id, is_oa, oa_status, authors, n_authors, primary_field, fields_all. These attributions preserve the main data while converting the data into a table that can easily be analyzed. Works with no concept, no venue, or no publication date were handled safely, so they wouldn't crash the code, but were retained.

```
Task 2.3: Create and Save the DataFrame
```

```
import pandas as pd
flattened = [flatten_work(work) for work in works]
df = pd.DataFrame(flattened)
df.head()
```

	openalex_id	doi	title	publication_year	publication_date	type	cited_by_count	venue
0	https://openalex.org/W2963002078	https://doi.org/10.1093/ptep/ptac097	Review of Particle Physics	2022	2022-08-01	article	6045	Progress of Theoretical and Experimental Physics
1	https://openalex.org/W4401211033	https://doi.org/10.1103/physrevd.110.030001	Review of Particle Physics	2024	2024-08-01	article	3585	Physical review, D/Physical review, D.

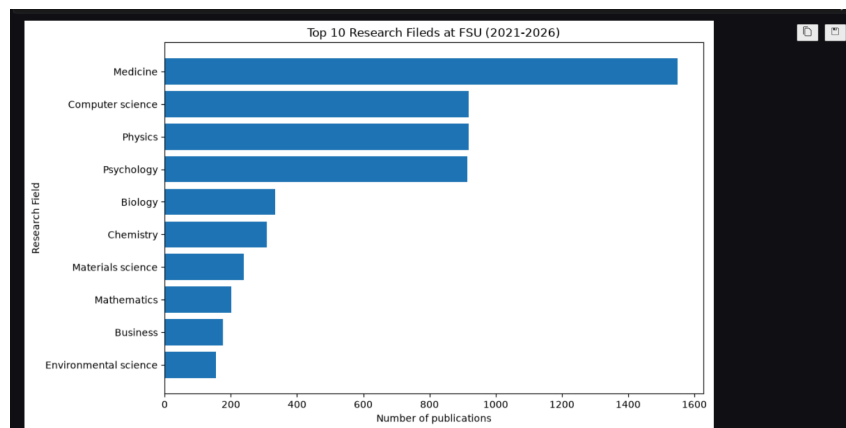
```
df.to_csv("fsu_works_2021_2026.csv", index=False)
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
```

Rows: 27733
Columns: 15

Statistics and charts:

Published counts increased over the years, with 2026 needing to be interpreted carefully due to it not being over yet and thus being incomplete.

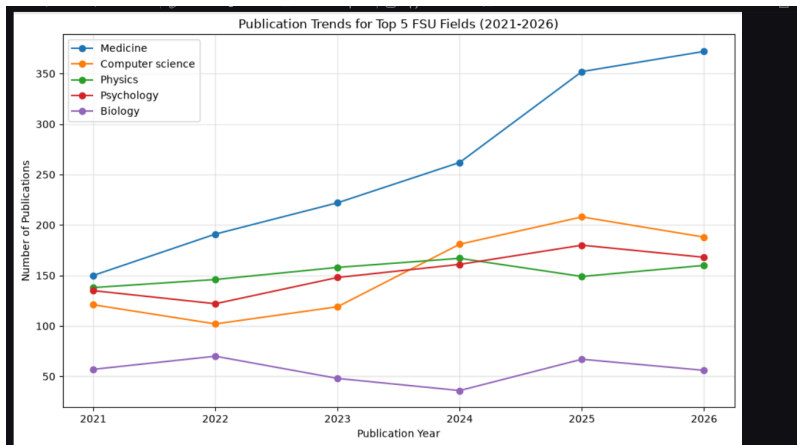
Top 10 bar chart:



Medicine clearly has the highest amount of publications, followed by a nearly tied comp sci, physics, and psych, with it steadily decreasing until the lowest, environmental science. However,

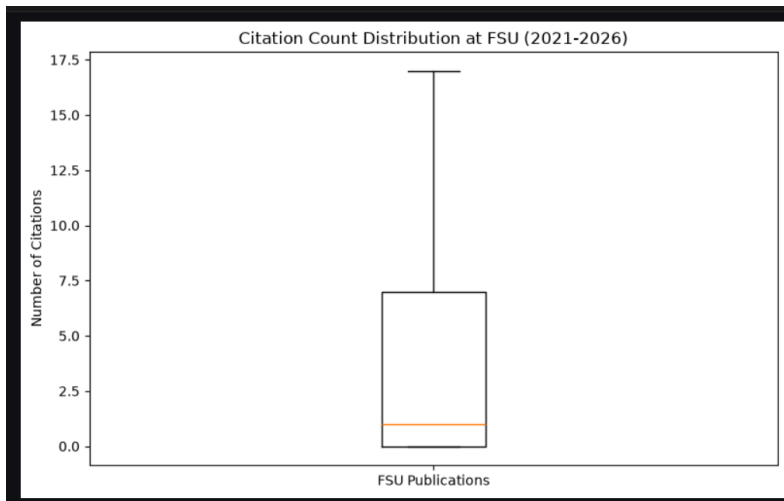
this chart only shows the top 10 fields, so it hides the remaining fields and doesnt show how it changes over time

Line chart:



The line chart compares publication trends in the five most common research fields at FSU, letting the reader easily see changes in trends within each fields. However, it only compares the top 5 fields, and since 2026 isnt over yet, it shouldnt be interpreted as final annual totals

Box plot:



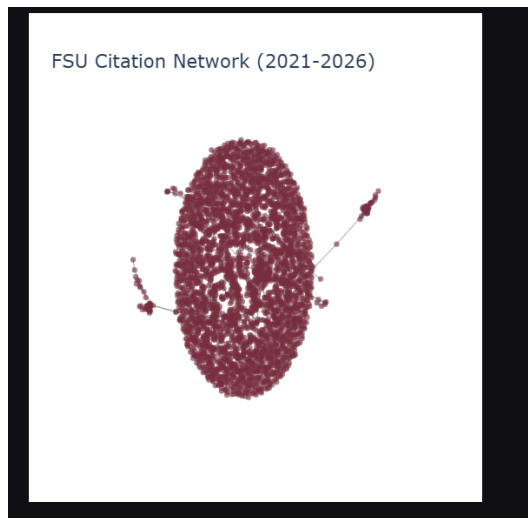
The distribution is strongly skewed with most publications recieving 7 or less, with the median being 1, however, the maximum is very high. The chart hides extreme outliers so the dsitribution is easier to see, so some highly cited papers arent visible

Citation network:

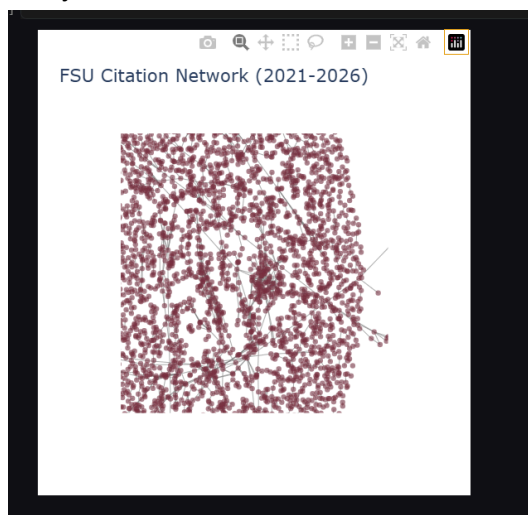
I took a random sample of 5000 papers without replacement from over 27,000 works, using random seed 42 to allow the sample to be reproducible. By sampling, it reduced the amount of requests needed while still showing the relationships

The sample created over 200,000 links in total, and after removing all where either the cited and the citing paper were not from FSU, it left over 3400 links left, or less than 2%. Clearly, most research outside of FSU's database.

Plotly full view



Plotly zoomed view



The citation network represents relationships between FSU affiliated publications, each node represents a paper and each edge represents a citation. It helps show the relationship between groups of publications and citations. The dense regions, however, make it hard to see individual

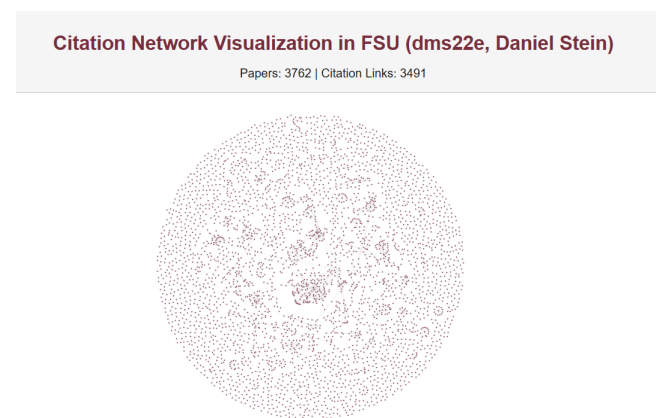
relationships without zooming in. Also, since it is based on only 5000 citing papers, it doesn't represent the full dataset

Interactive D3 Visualization:

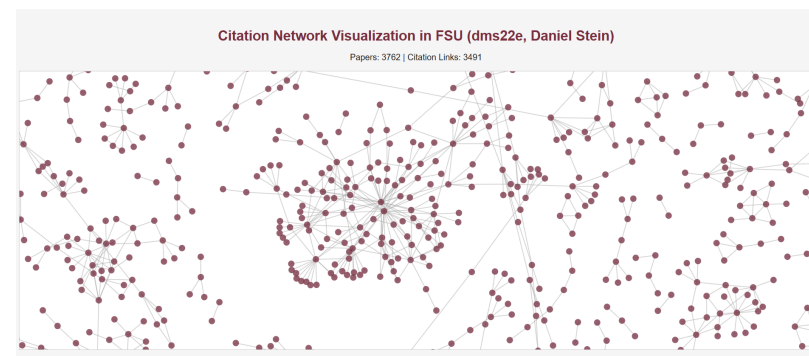
I separated the application into a React/Vite frontend and a Flask backend. The backend reads the citation CSV and publication metadata, aggregates links, builds the node and link structure, and exposes them through `/api/network`, as well as exposing `/api/stats` for network-size info. The frontend server can access the Flask API, as the frontend fetches the network JSON and passes it to a D3 module, which lays out the graphs. All the links and nodes are placed inside a single `<g>` element so D3 zoom and pan can change the entire graph.

<https://github.com/DanielStein15/assignment-1-dms22e-DanielStein>

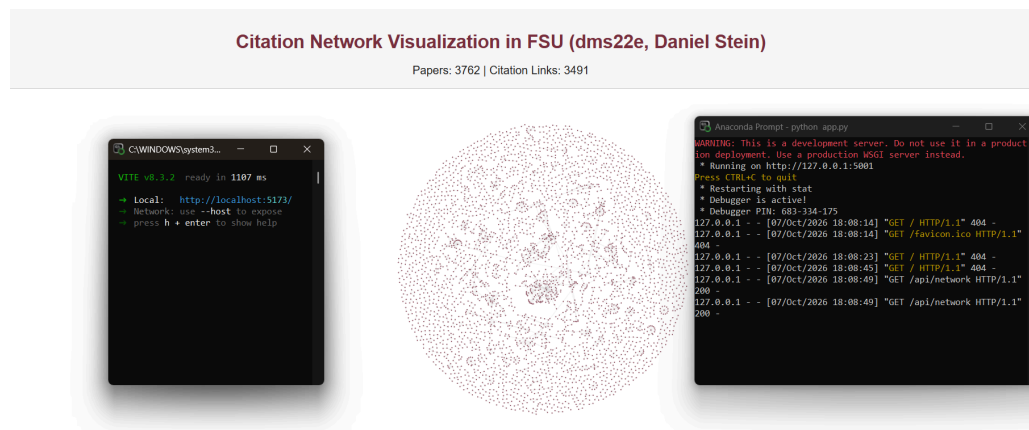
Full interactive network



Zoomed in network



Frontend and backend running together



Network analysis:

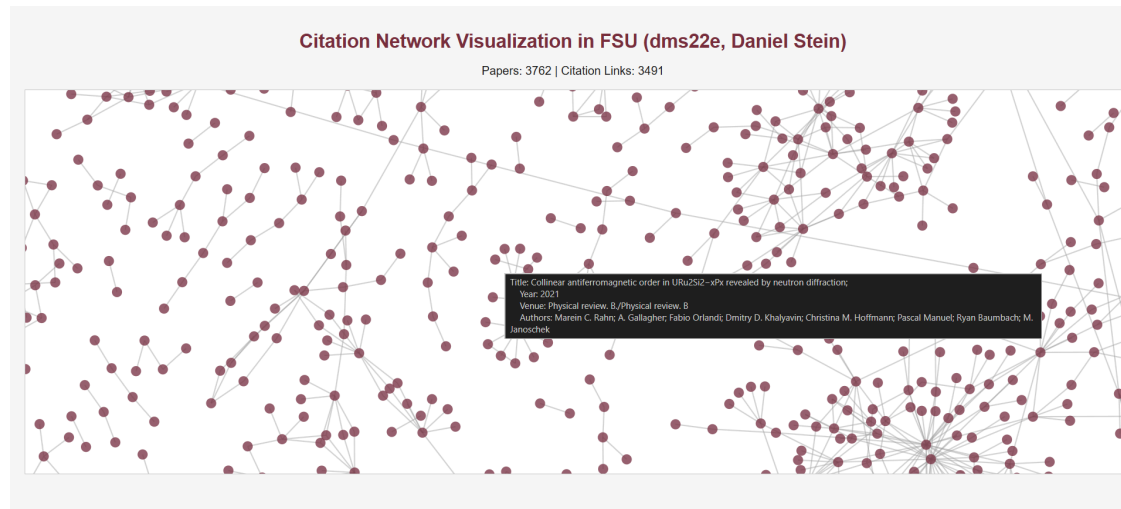
The citation network has 3762 unique papers (nodes), and 3491 unique links (edges) after removing the duplicates. I used a directed network, since citations have a direction - one cites another. Each pair is counted as one edge

Top 10 most cited papers:

	title	publication year	venue	citation count
257	Electron and photon reconstruction and identif...	2021	Journal of Instrumentation	29
380	Precision luminosity measurement in proton-pro...	2021	The European Physical Journal C	28
142	The CMS Statistical Analysis and Combination T...	2024	Computing and Software for Big Science	15
885	Review of Particle Physics	2022	Progress of Theoretical and Experimental Physics	14
19	Is personality associated with dementia risk? ...	2021	Ageing Research Reviews	11
724	Sense of purpose in life and concurrent loneli...	2022	Journal of Affective Disorders	9
281	Review of Particle Physics	2024	Physical review. D/Physical review. D.	9
442	A Surface Discharge Model for the Design of Su...	2022	IEEE Journal of Emerging and Selected Topics i...	6
1526	Nutraceuticals in Bulk and Dosage Forms: Analy...	2021	Molecular Pharmaceutics	6
725	Sense of meaning and purpose in life and risk ...	2022	Archives of Gerontology and Geriatrics	6

The ten most cited pages are listed above, ranked by in-degree. This is different from OpenAlex's `cited_by_count` which counts citations from papers worldwide. Since this is just FSU-internal links, the citation counts are more limited, and shouldn't be interpreted as each paper's worldwide citation. The highest cited papers are in physics related subjects, suggesting that internal FSU network has physics as some of their most prominent, but also has many other fields that are well cited.

Bonus work:



Shows the title, year, venue, authors when you hover over it